Fatgezim “Zim” Bela

Software & Data Engineer | Systems Integration | Technical Leadership

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fatgezimb.github.io/fatgezim-portfolio

Professional Summary

Python-focused engineer and Founder/CTO with experience in predictive algorithms, data workflows and application architecture. Programming since 2013, beginning with Python automation, analysis and web applications. Independently developed an algorithm under restricted-data access constraints and led hands-on product development from business requirements through implementation.

Engineering Skills

Data and algorithms Python, SQL, data analysis, predictive modeling, validation and automation

Application development JavaScript, TypeScript, React, Next.js, APIs and systems integration

Architecture and delivery Requirements analysis, data modeling, authentication, Git and testing

Relevant Engineering and Clinical Experience

Data Engineer **October 2022 - January 2023**

The McVay Company, Inc.

- Independently designed and implemented a custom adaptation of Dijkstra’s shortest-path algorithm for large-scale health-related data analysis on a Department of Defense project.
- Created synthetic development and test datasets and established initial validation without access to restricted operational data.
- Transferred the solution for team testing on operational data; colleagues reported processing millions of data points in seconds and narrowing results to approximately 12 records for manual review.

Founder and Chief Technology Officer **2018 - Present**

Create 13 Group

- Co-founded a technology and business development company; lead technical requirements, architecture, technology selection and hands-on implementation.
- Led architecture and development of a consumer-brand commerce platform in 2026, connecting responsive shopping interfaces, product and inventory management, cart workflows and administration.
- Defined database, authentication and payment-integration architecture using TypeScript, React, Next.js, Vercel and Stripe; design workflow automations and external-service integrations.

Founder and Chief Technology Officer **2026 - Present**

NeuroPath LLC

- Co-build Bela Behavior Data Lab with Meili Bela, turning appointment exports into browser-local dashboards, follow-up views and reports.
- Translate clinical and learning requirements into structured records, review workflows and operational software for a behavioral-health practice.

Board Certified Behavior Analyst **2025 - Present**

Clinical practice

- Use behavioral data to guide individualized programming, caregiver collaboration and supervised RBT implementation.

Selected Engineering Projects

Clinical Assessment Portal

Private implementation tested with synthetic records | React, TypeScript, Express, Firebase

- Implemented practice-scoped authorization and record ownership, with explicit edit permissions.
- Added version checks, clinician review and signatures, and server-generated PDF and DOCX reports.
- Integrated Firebase authentication and Firestore persistence with Zod validation and Vitest testing.

Relay Commissioning Readiness **View project**

Independent engineering prototype with fictional commissioning records | React, JavaScript

- Modeled acceptance as a dependency chain, tying evidence to the current asset revision and prerequisite event IDs.
- Derived readiness from the complete chain; revision changes and replacement test records invalidate dependent approvals.
- Covered blocked acceptance, revision propagation and persistence recovery with 10 domain tests; added JSON turnover exports.

Grid Forecast Lab **View project**

Historical demand forecasting benchmark | Python, scikit-learn, React

- Built a day-ahead forecast pipeline with expanding chronological validation, separate interval calibration and held-out 2017 evaluation.
- Measured 1,854 MW mean absolute error over 8,732 test hours, 14.4% below the prior-day baseline.
- Extended a credited Rob Mulla Kaggle exercise with a reproducible pipeline, executed notebook, tests and interactive forecast replay.

Research

Computational Neuroscience Research | George Mason University **Poster**

Co-author with Sumit Nanda and Giorgio A. Ascoli on Tracing Neurons from the Fruit Fly Larva.
Contributed to a collaborative workflow documenting image-channel processing, stitching, neuronal tracing and vectorized-tree reconstruction.

Methods: Fiji/ImageJ, Vaa3D and NeuTube

Education

Doctor of Medicine Candidate | Ross University School of Medicine | Expected 2028

M.Ed. in Special Education | George Mason University | 2020

Graduate Certificate in Applied Behavior Analysis | George Mason University | 2020

B.S. in Neuroscience | George Mason University | 2017

Professional Credentials

Board Certified Behavior Analyst (BCBA).

Licensed Behavior Analyst in Virginia, North Carolina and New York.